VIP Syllabus Spring 2026
VIP Team: AIDA3 Research

VIP 17910, 17912, 27920, 37920, 47920, 47921, 47922
https://engineering.purdue.edu/vip/

Course Information

- **CRN:** see https://engineering.purdue.edu/vip/register
- **Meeting day(s) and time and Instructional Modality**
  - **Lecture**: Hybrid (Professional Development)
  - **Lab (Team): Instruction Modality**: Face-to-Face, unless defined otherwise
  - **Lab (Team) Meeting day(s) and times**: Wednesdays 5:30 pm - 7:20 pm
  - **Lab Meeting Location/Information**: ME 1012
  - **Dates of instruction:** January 12-May 9th (Finals are May 4-9). Note: Course will not meet January 19th, March 16-21st.
- **Course credit hours:**
  - Two credit hours: VIP 27920, 37920, 47920, 47921, 47922
  - One credit hour: VIP 17910, 17912
  - Zero credit hour: VIP 37900

Course Description

This VIP research program develops machine-learning models for fixed-wing UAVs and participates in national and international competitions—both virtual and physical—to advance autonomous aviation. Students will organize themselves in smaller teams that fall in the program's core research thrusts, including (1) perception, navigation, and control, as well as (2) digital twin and simulation, but also (3) have the opportunity to work on projects outside of these thrusts in the broader area of AI for autonomous aviation (e.g. human autonomy teaming). Teams are encouraged to explore and establish novel methodologies, such as nonlinear model predictive control, reinforcement learning, physics-informed ML, generative AI (LLMs), or embodied AI. Typical activities include software development in Python/C++/ROS 2, training and benchmarking models, building and testing simulation data pipelines, and participating in competitions. Over multiple semesters, teams aim to compete in different national and international competition events and create tools, datasets, and publications for autonomous aviation.

VIP Team AIDA3 Research Personnel

VIP Team AIDA3 Research Role	Name	Office	Office Hours	Email
Instructor/Faculty Mentor	Dr. Sabine Brunswicker	MRGN 148	By appointment	sbrunswi@purdue.edu
Instructor Mentor	Worawis Sribunma	NISW 118A	By appointment	wsribunm@purdue.edu

VIP Program Personnel

VIP Position	Name	Office	Email
Director/PD Instructor	Dr. Carla Zoltowski	ARMS 1217	cbz@purdue.edu
Assistant Director/ PD Instructor, WL	Cyndi Lynch	ARMS 1219	clynch@purdue.edu
Assistant Director/ PD Instructor, Indianapolis	Dr. Steve Higbee	SL 220D	shigbee@purdue.edu

Senior Program Manager, WL	Dr. Andrea Tener	ARMS 1236	atener@purdue.edu
Senior Program Manager, VIP/EPICS, Indianapolis	Matthew Johnson	ET 215A	john3892@purdue.edu
Lab Manager, VIP/Honors, WL	Rick Womack	LMBS 2277	rwomack@purdue.edu
Director of Innovation Hall Maker Space, Indianapolis	Dr. Andy Tomaschke	IO 112	atomasch@purdue.edu

Office hours for all: Email to schedule
Emails: For WL specific questions: VIP_WL@purdue.edu; For Indianapolis specific: VIP_Indy@purdue.edu
For general VIP questions: vip@purdue.edu

Learning Outcomes:

Students in VIP will make progress on each of the learning outcomes listed below:

- An ability to apply engineering design and innovate software solutions to advance autonomous aviation that meets the specified needs of the design requirements, with consideration of safety, performance, robustness, and novelty.
- An ability to plan and conduct simulation and flight-test experiments (digital twin, HITL/SITL, indoor motion-capture testing), analyze telemetry and performance metrics (e.g., lap time, tracking error, constraint violations, stability margins), and use engineering judgment to improve autonomy performance, while considering relevant operational constraints and broader impacts.
- An ability to identify, formulate, and solve complex engineering problems that arise in autonomous aviation competition settings (e.g., model uncertainty, disturbances (wind/turbulence), sensor noise, latency, actuator limits, and safety constraints) by applying principles of controls, dynamics, estimation, and software engineering.
- An ability to integrate autonomy software across a full pipeline (simulation infrastructure, APIs, ROS2/autopilot interfaces, configuration management, logging/scoring scripts), using reproducible workflows (version control, testing, documentation) and engineering best practices to deliver reliable competition-ready systems.
- An ability to function effectively on a multidisciplinary team to define roles, establish goals, plan sprints/test campaigns, manage interfaces, and deliver milestones while practicing collaboration norms used in flight-test and autonomy engineering (reviews, checklists, issue tracking, and post-test debriefs).
- An ability to communicate effectively with a range of audiences appropriate to this design experience in both a written report and an oral presentation.
- An ability to acquire and apply new knowledge as needed, using appropriate learning strategies to complete the engineering design experience associated with this course.
- An ability to recognize ethical and professional responsibilities associated with this engineering design experience and make informed judgments that must consider the impact of the product of the engineering design experience

Software Tools

The software tools will vary based on the project. However, in general, the course and teams will use the following software tools:

- Brightspace: Within Brightspace, you will have access to course announcements, schedules, assignments, grades, feedback, and course resources.
    - For Overall Course Announcements and Professional Development (PD) opportunities: Spring 2026 VIP AIDA3 Research PD – Merge

    - For lab/team: Spring 2026 VIP AIDA3 Research – Merge
- MS Teams, WebEx, Zoom, Slack and/or other conferencing/meeting/messaging platforms
- CATME: You will use CATME to submit information used for Peer & Team Evaluations (Link to CATME).
- MS Office: Word, Excel, and PowerPoint.
- Adobe: PDF.
- Qualtrics: https://purdue.qualtrics.com/
- Simplicity: To register for in-person PD activities: https://engineering-purdue-csm.symplicity.com/
- Python/C++
- Robot Operating System(ROS 2)

Attendance Policy

Within the VIP course, you will be working in teams on a project. Your project work, both individually and as part of the team, are the most significant aspects of the course. Thus, you are expected to participate in all scheduled VIP team meetings (the lab portion of your VIP course), as well as any sub-team meetings. Meetings will be conducted via in-person or virtually on WebEx, Zoom, or other tool(s) as designated by your VIP team mentor and/or decided by you and your project team. This expectation aligns with Purdue's academic regulations regarding attendance (see Academic Regulations: Attendance and Office of the Dean of Students: Class Absences posted in Brightspace under "University Policies and Statements."), which states that students are expected to be present for every meeting of the classes in which they are enrolled. When conflicts or absences can be anticipated, such as for many University-sponsored activities and religious observations, you should inform the team mentors (instructors) and teammates of the situation as far in advance as possible. Furthermore, you are responsible for knowing what occurred in that meeting (typically by discussing it with other team members) and how you can contribute to the project and team until the next meeting. An excused absence does not relieve you of that responsibility.

For unanticipated or emergency absences when advance notification to is not possible, contact your team mentors (instructors) and teammates. For cases that fall under **excused absence regulations**, you or your representative should contact or go to the Office of the Dean of Students (ODOS) website to complete appropriate forms for instructor notification. Under academic regulations, excused absences may be granted by ODOS for cases of grief/bereavement, military service, jury duty, parenting leave, or certain types of medical care.

Grading

In this class, grades reflect the sum of your achievement of learning outcomes throughout the semester. Your grade is based on three areas (Accomplishments & Effort, Documentation, and Teamwork & Interactions), along with seven requirements, and is assessed according to your course level and major. Non-zero credit VIP courses must be taken in "Regular Grade" mode; registration as "Pass/No Pass Option" mode is not allowed. Students registered for the 0-credit version of VIP are graded as "Satisfactory / Unsatisfactory". The regular grading guidelines are as follows; please note that "+" or "-" grades may be given if the assessment falls above or below, respectively, the stated guideline, in alignment with Purdue's standard undergraduate grading scale (A: 93 – 100%, A-: 90 – 92.99%, B+: 87–89.99%, B: 83–86.99%, etc.).

Grade	Description
A	Overall, the student's accomplishments and effort, documentation, and teamwork and interactions are excellent. All of the seven (7) requirements have been satisfied.

B	Overall, the student's accomplishments and effort, documentation, and teamwork and interactions are good. Six (6) of the seven (7) requirements have been satisfied.
C	Overall, the student's accomplishments and effort, documentation, and teamwork and interactions are adequate. Five (5) of the seven (7) requirements have been satisfied.
D	Overall, the student's accomplishments and effort, documentation, and teamwork and interactions are marginal. More than two of the seven (7) requirements are missing.
F	Overall, the student's accomplishments and effort, documentation, and teamwork and interactions are unacceptable. More than three of the seven (7) requirements are missing.

Seven Requirements and Due Dates (Course Schedule)

As part of the assessment of the above, each student is required to:

1. Contribute appropriately to the project and document those accomplishments and contributions to the project and team in format as required by your project mentor(s). Specific requirements for your project will be posted on your team's Brightspace.
2. Contribute as appropriate to project documentation, presentations, publications, and/or poster. Specific requirements for your project will be posted on your team's Brightspace.
3. Submit Professional Development plan by Friday, January 30th at 11:59 pm and complete PD activities, including the Welcome to VIP lecture.
4. Complete Mid-semester Individual Performance Evaluation (IPE) by Friday, February 27th at 11:59 pm in Brightspace.
5. Complete Final Individual Performance Evaluation (IPE) & PD Reflection (PDR) by Friday, May 1st at 11:59 pm in Brightspace.
6. Complete mid-semester and final peer evaluation of team members in CATME (mid-semester due Friday, February 27th at 11:59 pm and final peer evaluations due Friday, May 1st at 11:59 pm.
7. Complete final Purdue course evaluation and submit a screen shot of completion to Brightspace (due Friday, May 1st at 11:59 pm).

Performance assessment will be done once at mid-semester and again at the completion of the semester using the criteria that follows. The mid-semester assessment is advisory (formative).

Accomplishments and Effort:

__ Quantity of project accomplishments
__ Quality of project accomplishments
__ Initiative
__ Work ethic
__ Ability to overcome project setbacks
__ Learning needed for the project
__ Focuses effort on achieving goals
__ Manages time and tasks well
__ Overall

Documentation:

__ Individual documentation (quality/quantity)
__ Contributions to team documentation (quality/quantity)
__ Contributions to team poster/presentations
__ Use of appropriate tools (e.g., Git)
__ Overall

Teamwork and Interactions:

__ Participates fully in team (lab) meetings
__ Participates fully in sub-team meetings
__ Contributes fair share of team's work with acceptable quality
__ Communicates clearly. Shares information with teammates
__ Respects and responds to feedback from teammates

__ Keeps commitments and completes assignments on time
__ Listens to teammates and respects their contributions
__ As appropriate, involves and assists others in efforts
__ Demonstrates leadership and/or project management skills
__ **Overall**

Professional Development (PD) Plan

To support your success on your project and your team, and overall, in your career and your life, you will create an individualized Professional Development Plan. For all students *except* for those enrolled in VIP 17910, 17911 or 17912, the plan is expected to incorporate ten (10) Professional Development (PD) activities that can be completed throughout the semester and include the three required activities: the Welcome to VIP "lecture" and the two Communication activities. For students enrolled in VIP 17910, 17911 and 17912, the PD Plan only needs to include the *three* required activities. Most VIP students on West Lafayette teams typically participate in the Purdue Undergraduate Research Conference through the Office of Undergraduate Research (OUR) to complete the abstract and presentation requirements. Although students on VIP teams in Indianapolis can also participate in the OUR conference, they are expected to participate in the EPICS/VIP Poster session in Indy as well. Other activities designated and approved by the instructor can be used to satisfy this requirement.

Track/Topic*	Activity	Organizing unit	Date (of activity or expected participation)
1. PD (Required)	Welcome to VIP	VIP	Complete by Friday, January 30th at 11:59 pm
2. Communication (Required) Complete instructor approved option for your project	Research Conference (West Lafayette) – Abstract submission	OUR	Abstract due: March 2nd at 11:59 pm See Purdue Undergraduate Research Conference Website
	Student Innovation & Excellence Showcase (Indianapolis) – Abstract Submission	OUR	Abstracts due March 25 at 11:59 pm. Purdue Undergraduate Research Conference Website
	Instructor designated activity		
3. Communication (Required) Complete instructor approved option for your project	Research Conference (West Lafayette) – Poster Session/ Research Talk	OUR	Poster session/research talks: Friday, April 14 & Saturday, April 18th See Purdue Undergraduate Research Conference Website
	Student Innovation & Excellence Showcase (Indianapolis) – Poster Session	Purdue University	Poster Sessions I & II: April 30th See Purdue Innovation & Excellence Showcase Website
	Instructor designated activity		

The remaining seven activities are to be chosen by the student with guidance from their VIP mentor(s), depending on their interest and what would be most beneficial for their work on a specific team. Please see the Team Expectations Section below for specific suggestions for this team.

The VIP PD Brightspace site hosts a variety of Professional Development videos and information on interactive workshops and live presentations that will be offered this semester. Students should browse the materials and talk with their mentors during the first week of the semester to create their individual Professional Development (PD) Plan and to ensure that project and lab requirements are met. The PD plan should be submitted on the team Brightspace by **Friday, January 30th**. Students may update their plans as needed during the semester without resubmitting to Brightspace.

Students will be expected to document their participation in each of the activities via the assignments posted in their team Brightspace, as well as in the Midsemester Individual Performance Evaluation (IPE) and the Final IPE & PD Reflection assignments.

Senior Design Students (VIP 47921/47922)

In addition to the **Seven Requirements and Due Dates (Course Schedule)** stated above, senior design students must complete the following documents (templates are posted in Brightspace):

- VIP Senior Design Project Proposal: Must be completed **individually** by each senior design student during the first semester of Senior Design to ensure he/she has an appropriate project and role. This is to be submitted at the mid-semester and final evaluations during the first senior design semester.
- VIP Senior Design Project Description: Must be completed during the second semester of Senior Design by each project team. **One version per project team** must be submitted at the mid-semester and final evaluations during the second senior design semester. Please see note below on format of final submission for project teams with second semester senior design students.
- VIP Senior Design Reflection document: The Senior Design Reflection document is submitted at the final evaluation for both senior design semesters **in addition to** the Individual Performance Evaluation rubrics.
- The Senior Design Project Proposal/Description and the Senior Design Reflection documents will be used by the instructors(s) and VIP admin in conjunction with the project and course requirements to approve the satisfaction of the course outcomes and in determining the course grade.
- **Format of final submission for 2nd semester students**: For the final submission, the project team should submit a compiled document which contains the team Project Description and the Senior Design Reflection for all the 2nd semester students on the project.

Grading for Senior Design Students

Senior design students in VIP are expected to make progress on all course outcomes in the first semester and demonstrate competency by the end of the second semester. In addition, students make significant progress towards the project final deliverable each semester. For the first semester, students are typically expected to have completed prototypes and proof of concepts of the project, and the outcomes will be evaluated accordingly. For the final (second) semester, the outcomes will be evaluated based on the final state of the student project, which typically consists of a fully completed system or a well-developed subsystem. See the Senior Design ABET Outcome Rubric Metric for more details.

Each outcome will be rated on a scale from 1 to 4, where 4 is “Excellent”, 3 is “Good”, 2 is “Adequate/Acceptable”, and 1 is “Inadequate/Unacceptable” as appropriate for first or second semester expectations. See the Senior Design ABET Outcome Rubric Metric for more details.		**Weight**
i.	An ability to apply engineering design to create a product that meets the specified needs of this engineering design experience with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors.	30%
ii.	An ability to develop and conduct experimentation, analyze and interpret data, and use engineering judgment to draw conclusions related to the development of the product of this engineering design experience.	15%
iii.	An ability to identify, formulate, and solve complex engineering problems arising from this engineering design experience by applying principles of engineering, science, and mathematics.	15%
iv.	An ability to function effectively on a team whose members together provide leadership, create a collaborative environment, establish goals, plan tasks, and meet objectives associated with this design experience.	10%
v.	An ability to communicate effectively with a range of audiences appropriate to this design experience in both a written report and oral presentation.	10%
vi.	An ability to acquire and apply new knowledge as needed, using appropriate learning strategies to complete the engineering design experience associated with this course.	10%
vii.	An ability to recognize ethical and professional responsibilities associated with this engineering design experience and make informed judgments which must consider the impact of the product of this engineering design experience in global, economic, environmental, and societal contexts.	10%

AIDA3 Research Areas:

AIDA³ research focuses on turning advanced AI and uncrewed aircraft technologies into deployable, safety-aware capabilities for modern aviation.

AIDA3 research is organized into 5 research thrusts:



In 2026, our work is organized into six projects spanning three primary research areas:

1.) **LLM End-to-End Mission Planning**

Building AI copilots and planning stacks that can translate intent into operationally valid, constraint-aware missions.

1. LLM ATC Chatbot
2. Multi-model Translation & Reasoning Engine
3. Neuro-symbolic AI for Dynamic and Verifiable Mission-Planning (STL)

2.) **Health Monitoring**

Enabling Resilient operations through data-driven monitoring, anomaly detection, and model-aware diagnosis

4. Counterfactual Reasoning Engine Health Monitoring
5. System Identification for Aircraft Health Monitoring

3.) **Digital Twin and Mixed-Reality Testbed for Accelerated Lab-to-Life**

Reducing risk and development time by connecting high-fidelity simulation to scaled mixed-reality flight testing to real-world flight testing

6. Digital Twin and Mixed-Reality Testing

Additional Resources:

AIDA3's webpage: https://www.purdue.edu/computes/aida3/

LLM Route Planning Demo: https://youtu.be/OmfWZqO04Ug?si=GayIvQH28lnMGxQ8

AIDA3-Windracers ULTRA Flight: https://youtu.be/gs_ZTh7p5bM?si=OESjRScSg7AqNnfd

Autonomous Drone Racing Competition Details:

In Fall 2025, AIDA3 hosted the Autonomous Fixed-Wing UAS Pylon Racing competition, where teams are tasked to develop autonomy tools and Guidance, Navigation, and Control (GNC) framework to navigate a fixed-wing UAS around a marked pylon course autonomously. Teams submit their solutions to our cloud-based digital twin simulation platform, hosted in collaboration with Purdue Rosen Center for Advanced Computing (RCAC), where teams can view their performances in our real-time leaderboard system. Finalists are selected to deploy their solution onto a physical fixed-wing UAS and fly their solution within Purdue UAS Research and Test facility (PURT), the world's largest indoor motion-capture facility. In Spring 2026, we will scale the competition up

Competition Webpage: https://pylon-racing.geddes.rcac.purdue.edu/

December 2025 Competition Video: https://youtu.be/Cp59R1KoB8c?si=n_r4lK2GlD2t9bix

AIDA3-Specific Research and Competition Schedule

Week	Date	Design Activity Phase	Competition Phase/Key Dates (participation is encouraged)	*Student assignment and deliverables (due before class meeting)*
1	1/14/2026	Problem Formulation/Project Description		
2	1/21/2026	Problem Formulation/Project Description		*Team selection*
3	1/28/2026	Problem Formulation/Project Description	Kick-off and announcement	*Research proposal (Research question, technical plan (model, data/experiment/validation), and timeline - roadmap, tools/framework to be used); based on template (1 page max)*
4	2/4/2026	Data Collection and Modeling	Info Session 1: Competition Requirements/Announcement	
5	2/11/2026	Data Collection and Modeling		
6	2/18/2026	Data Collection and Modeling		
7	2/25/2026	Data Collection and Modeling	Info Session 2: Demo/Example solution	*March 2, abstract submission deadline: Spring undergraduate research conference*
8	3/4/2026	Simulation, Testing, Validation		
9	3/11/2026	Simulation, Testing, Validation	Info Session 3 (tentative)	*Research progress report*
10	3/18/2026	Spring break/no class		
11	3/25/2026	Simulation, Testing, Validation	March 28, 2026 -- Virtual Solution Submission	
12	4/1/2026	Demo, Documentation, and Finalization		
13	4/8/2026	Demo, Documentation, and		

		Finalization		
14	4/15/2026	Demo, Documentation, and Finalization	April 18, 2026 -- In-person Competition & Award Ceremony	*April 14, Poster Session*
15	4/22/2026	Data Analysis and Reporting		
16	4/29/2026	Data Analysis and Reporting		*End-of-semester report*

Academic Integrity and Professional Responsibility

Academic integrity is one of the highest values that Purdue University holds. The VIP Program expects every member of the Purdue community to adhere to the Purdue Honor Pledge ("As a boilermaker pursuing academic excellence, I pledge to be honest and true in all that I do. Accountable together - we are Purdue.") and practice honorable, ethical, and professional behavior both inside and outside the classroom. In VIP, students are encouraged to work together and share information. When indicated, students and teams are allowed to modify previous versions of documents to be submitted for the current assignment. However, it is unacceptable for students to claim individual work that is not their own or to use sources, including generative AI, without appropriate citation. It is also unacceptable for students to misrepresent information to their instructional staff, their team, and/or their client. In addition, misuse of VIP resources is considered dishonest. At the instructor's discretion, instances of academic dishonesty will result in a reduced score, a zero score, or a failing grade for the course. All occurrences of academic dishonesty will be reported to the Office of Students Rights and Responsibilities (OSSR) and the students' respective schools. If there is any question as to whether a given action might be construed as academic dishonesty, please see the instructor or the teaching assistant before you engage in any such action. Details and resources are outlined on the Office of Student Rights and Responsibilities' Academic Integrity and You: Undergraduate Edition webpage.

Guidance on Generative AI

The appropriate use of AI tools is project and task specific. Please see specific guidance of allowable and appropriate use of AI tools for your project and task within your team's Brightspace and discuss with your VIP instructors and mentors if you have any questions regarding its use. However, use of AI tools to write reflections on individual learning is not allowed. For tasks that are allowed, you are required to document your usage of the AI tools and include details of how it was used within the project or task.

Lab and Facilities Standard Operating and Safety Procedures

The VIP Suite in West Lafayette is available for schedule team (lab) meetings, and for other VIP related meetings (see "Reserving VIP Lab Rooms on the Resources page (https://engineering.purdue.edu/VIP/resources). Teams using their mentor's research lab facilities, equipment, or resources are expected to follow the associated standard operating and safety procedures, including completing appropriate safety training. If applicable, more information will be posted on the team Brightspace.

Nondiscrimination Statement

Purdue University and the VIP Program are committed to maintaining a community that recognizes and values the inherent worth and dignity of every person; fosters tolerance, sensitivity, understanding, and

mutual respect among its members; and encourages each individual to strive to reach their potential. For details, see Purdue's full Nondiscrimination Policy Statement.

Accessibility

Purdue University strives to make learning experiences accessible to all participants. If you anticipate or experience physical or academic barriers based on disability, you are encouraged to contact the Disability Resource Center at: drc@purdue.edu or by phone: 765-494-1247, as soon as possible. Specifically, every member of our course should be able to access, use, and learn from the materials we share. This includes all course-related digital content that you and I share in the course. This approach helps promote equal access for everyone at Purdue and is mandated federally by Title II of the Americans with Disabilities Act (ADA). We will work together to provide this access within our Brightspace course.

Mental Health/Wellness Statement

If you find yourself beginning to feel some stress, anxiety and/or feeling slightly overwhelmed, try Therapy Assistance Online (TAO), a web and app-based mental health resource available courtesy of Purdue Counseling and Psychological Services (CAPS)**.** TAO is available to all students at any time by creating an account on the TAO Connect website, or downloading the app from the App Store or Google Play. It offers free, confidential well-being resources through a self-guided program informed by psychotherapy research and strategies that may aid in overcoming anxiety, depression and other concerns. It provides accessible and effective resources including short videos, brief exercises, and self-reflection tools.

If you need support and information about options and resources, please contact or see the Office of the Dean of Students. Call 765-494-1747. Hours of operation are M-F, 8 a.m.- 5 p.m.

If you find yourself struggling to find a healthy balance between academics, social life, stress, etc., sign up for free one-on-one virtual or in-person sessions in West Lafayette with a Purdue Wellness Coach at RecWell. Student coaches can help you navigate through barriers and challenges toward your goals throughout the semester. Sign up is free and can be done on BoilerConnect. Students in Indianapolis will find support services curated on the Vice Provost for Student Life website.

If you're struggling and need mental health services: Purdue University is committed to advancing the mental health and well-being of its students. If you or someone you know is feeling overwhelmed, depressed, and/or in need of mental health support, services are available. For help, such individuals should contact Counseling and Psychological Services (CAPS) at 765-494-6995 during and after hours, on weekends and holidays, or by going to the CAPS offices in West Lafayette or Indianapolis.

Additional Resources for Engineering Students:

- **The Community, Assistance and Resources for Engineering Students (CARES) Hub** was created to build lifelong thriving Boilermaker engineers by developing awareness, well-being and community for students through educational outreach, peer mentoring, easy access to mental health resources and social connection. The CARES Hub, in Room 1261 of the Neil Armstrong Hall of Engineering (ARMS) that is designed for engineering students to relax, study and connect with fellow engineering students. In addition to the CARES Hub, CARES offers low-barrier, high-access support for students' well-being and mental health needs through on-site therapy, peer mentoring, wellness education and social activities designed to build connection and community. See the CARES Hub website for hours.

Basic Needs Program

If you are facing challenges securing basic needs such as food, housing, transportation, health services, or access to technology or childcare resources and believe this may affect your performance in the course, please contact the Office of the Dean of Students (ODOS) to help coordinate with community resources. These services vary by location. In **West Lafayette**, see the Basic Needs Program website, or email basicneeds@purdue.edu. To connect with a Student Support Generalist on the **Indianapolis** campus, contact them by phone at 765-495-7797 or email studentlifeindy@purdue.edu.

Emergency Preparation

In the event of a major campus emergency, course requirements, deadlines and grading percentages are subject to changes that may be necessitated by a revised semester calendar or other circumstances beyond the instructor's control. Relevant changes to this course will be posted onto the course website or can be obtained by contacting the instructors or TAs via email or phone. You are expected to read your @purdue.edu email on a frequent basis.

A link to Purdue's Information on Emergency Preparation and Planning is located on Brightspace under "University Policies and Statements." This website covers topics such as Severe Weather Guidance, Emergency Plans, and a place to sign up for the Emergency Warning Notification System.